The Turning Effect — Paper D

HARD · 50 MINUTES · 30 MARKS

NAME DATE SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Always convert centimetres to metres before you multiply, and give every moment the unit N m.
- The mark for each question is shown in square brackets on the right.
- No formula is given on this paper. If you cannot remember it, write down what you do know and work from there.
- Where a beam has its own weight, that weight acts at its centre.

SECTION 1 — SEVERAL FORCES AT ONCE

1. Four children sit on a see-saw. On the left: 250 N at 1.6 m from the pivot, and 200 N at 0.8 m. On the right: 300 N at 1.2 m, and a fourth child sitting 2.0 m from the pivot. The see-saw balances.

(a) Calculate the total anticlockwise moment. [2]

(b) Calculate the moment produced by the 300 N child. [1]

(c) Calculate the weight of the fourth child. [2]

2. A uniform beam is 4 m long and has a weight of 400 N. It is pivoted 1.5 m from its left-hand end. A load of 600 N hangs from the left-hand end.

(a) Calculate the moment of the 600 N load about the pivot. [1]

(b) Calculate the moment of the beam’s own weight about the pivot, and state its direction. [2]

(c) A downward force is applied at the right-hand end to balance the beam. Calculate its size. [2]

SECTION 2 — DESIGNING AND JUDGING

3. A worker can push down with a maximum force of 250 N. She needs to produce a moment of 400 N m to shift a paving slab with a crowbar.

(a) Calculate the shortest crowbar that would do the job. [2]

(b) Explain one practical problem with using a crowbar much longer than this. [2]

4. A student says: “The distance in the moment formula is just the distance from the pivot to where you push.” Explain what is missing from this statement, and when it matters. [3]

5. A tall bus travelling round a bend begins to tip. Explain, using moments and the idea of a pivot, why a bus with passengers on the upper deck tips more easily than an empty one. [4]

6. A wheelbarrow carries a load of 500 N whose weight acts 0.4 m from the wheel. The handles are 1.2 m from the wheel.

(a) Calculate the upward force needed at the handles to lift the barrow. [3]

(b) Explain why moving the load closer to the wheel reduces the force needed. [2]

7. A student claims: “Doubling the force always doubles the moment.” State whether this is true, and describe a situation in which doubling the force does not double the moment. [4]

End of paper. Check every answer has a unit or a form the question actually asked for.